

Investigating the influence of the step size  $\Delta t$ .

1. **Description of distance metrics.**
  2. **Bias: experiments and comments/questions.**
  3. **Monte Carlo error: experiments and comments/questions.**
  4. **Particle crossings.**
  5. **Iterations for the Newton solve.**
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**1. Description of distance metrics.** Consider the empirical spectral distribution

$$F_{n,M}(\lambda) = \frac{1}{NM} \sum_{j=1}^M \sum_{k=1}^N \mathbb{I}(\lambda_k^{n,j} \leq \lambda),$$

where  $N$  is the system dimension,  $M$  is the number of trials, and  $\lambda_k^{n,j}$  is recorded at time step  $n$  of trial  $j$ . This is compared to the exact (finite) distribution function

$$F_N(\lambda) = \frac{1}{N} \int_{-\infty}^{\lambda} K_N(x, x) dx$$

using both the Kolmogorov-Smirnov (KS) distance and the Wasserstein distance. The former is defined as the supremum over all distances,

$$D_{n,M}^{KS} := \sup_{\lambda} |F_{n,M}(\lambda) - F_N(\lambda)|,$$

and the one-dimensional  $p$ -Wasserstein distance is defined

$$D_{n,M}^W = \left( \int_0^1 |F_{n,M}^{-1}(u) - F_N^{-1}(u)|^p du \right)^{1/p}.$$

In the case  $p = 1$  a change of variables eliminates the need for the inverse CDF (quantile) [1]. For  $p > 1$  the quantiles are solved numerically. If not explicitly stated, assume  $p = 1$ .

**2. Bias.** Discretisation bias for all three schemes is investigated by calculating  $D_{n,M}$  at equilibrium. To be precise, we measure the distance at  $T = 8$  for schemes with  $M = 50$  independent trials and step sizes  $\Delta t = 2^{-k}$ ,  $k = 3, 4, \dots, 13$ .

Due to computational limitations we only investigate systems with dimension  $N = 20$  and  $N = 50$ . Here we focus on comparing the tamed and implicit schemes, but data for the standard Euler scheme is also available.

Figure 1 and Figure 2 describe the evolution of  $D_{n,M}$  at  $T = 8$  with  $k$ . The dotted vertical lines indicate the threshold at which  $2^{-k} = 1/N^2$ . Figure 3 describes the evolution of the Wasserstein distance for all  $0 \leq T \leq 8$ , again for each  $k$ .

**2b. Comments and questions.** Something to note is that, for all of these experiments, the number of trials was fixed at  $M = 50$ . Is this enough?

The distances for the implicit method mix seemingly for all choices of  $k$  here. Does this suggest  $\Delta t = \mathcal{O}(1)$  is a fine choice?

**3. Monte Carlo error.** Only a small number of experiments have been done so far; see Figure 4.

**4. Particle crossings.** One problem with the Euler and tamed schemes is that they do not preserve the non-colliding nature of DBM. We have performed a small numerical experiment that runs these schemes with  $M$  trials until some critical time  $T^*$ , and then determines how many of these trials observe particle crossings in the next step as a function of  $\Delta t$ . See Figure 5.

The nature (and implementation) of the implicit scheme means it always respects the non-colliding nature of the particles, and thus is not considered here. (For completeness it was considered in the experiment and agrees with this notion).

As expected, as  $N$  increases so does the average number of crossings. For  $\Delta t = 1/N^2$  no particles are observed to cross in this experiment, but this is by no means a guarantee - especially as  $N$  grows.

**5. Number of iterations per Newton solve.** The implicit scheme is currently implemented with Newton's method solving the minimisation problem at each step. It is expected that a larger  $\Delta t$  would have an adverse effect on the number of iterations needed for the Newton solve; recall that the Jacobian  $J = [\partial(\nabla g)_k / \partial x_\ell]$  has

$$J_{k\ell} = \begin{cases} -1/N(x_k - x_\ell)^2, & k \neq \ell, \\ \frac{1}{2}V''(x_k) + \frac{1}{\Delta t} + \frac{1}{N} \sum_{j \neq k} 1/(x_k - x_j)^2, & k = \ell. \end{cases}$$

The question is more about to what extent  $\Delta t$  has an influence. Note that the relative tolerance for the solve is currently set at  $10^{-6}$ , with a hard maximum of 20 iterations.

Figure 6 looks at the effect of  $\Delta t = 1/N$  and constant  $\Delta t = 0.1$  for systems of  $N \leq 500$ . We suggest the disadvantage here is negligible compared to the difference in total steps needed - especially because the Newton iteration uses conjugate gradient for  $\mathcal{O}(N^2)$  complexity instead of the naive  $\mathcal{O}(N^3)$  exact solve.

## References

- [1] SS Vallender. "Calculation of the Wasserstein distance between probability distributions on the line". In: *Theory of Probability & Its Applications* 18.4 (1974), pp. 784–786.

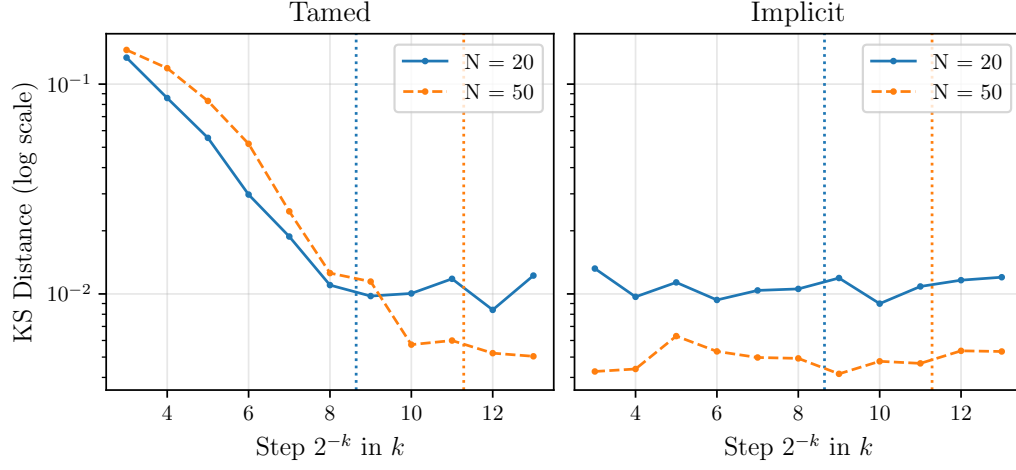


Figure 1: Evolution of the KS distance for the system at  $T = 8$  with  $\Delta t = 2^{-k}$ .

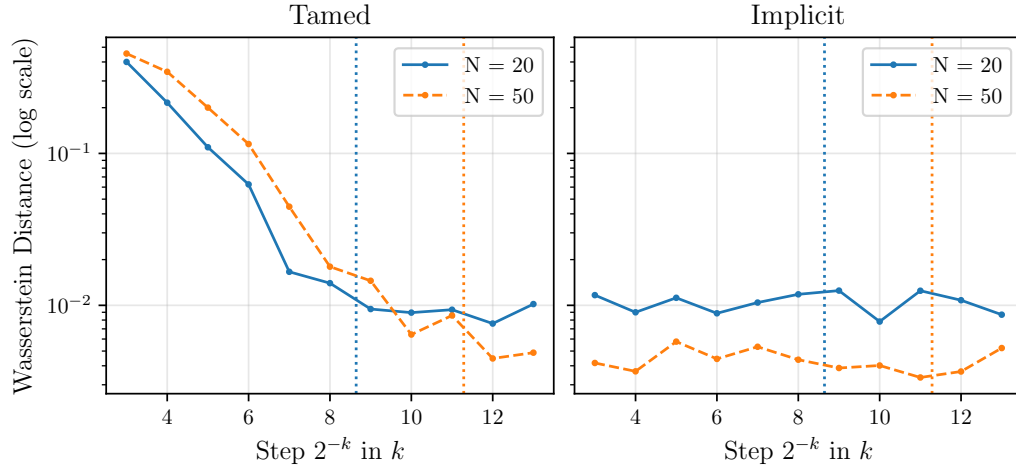


Figure 2: Evolution of the Wasserstein distance for the system at  $T = 8$  with  $\Delta t = 2^{-k}$ .

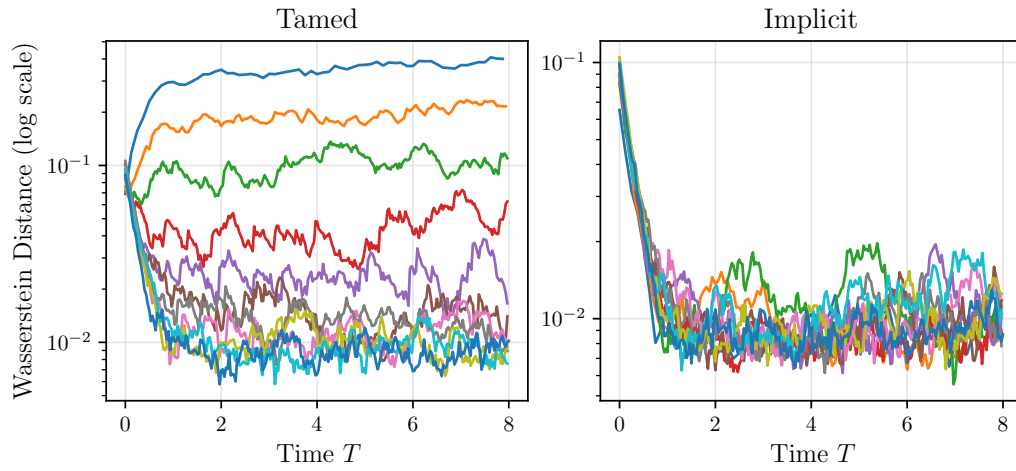


Figure 3: Evolution of the Wasserstein distance across all  $0 \leq T \leq 8$ , with each line representing an (indiscernible) choice of  $k$ .

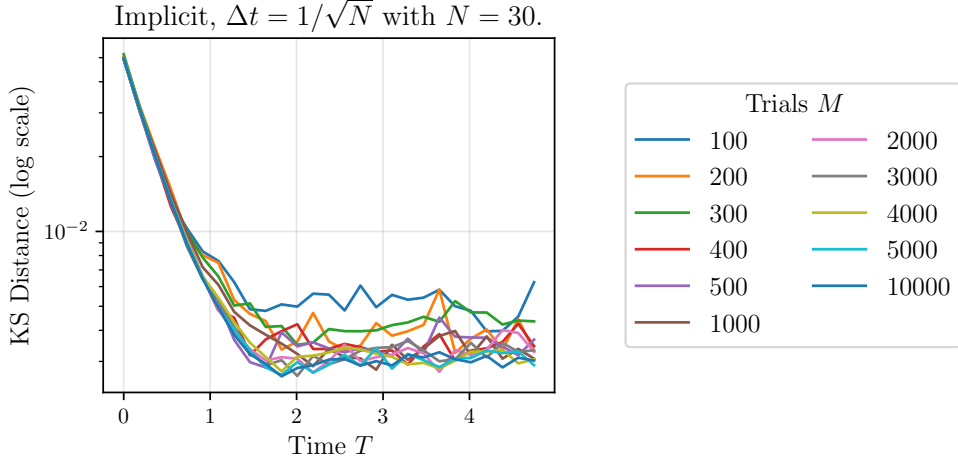


Figure 4: Evolution of the KS distance for a small system of  $N = 30$  with  $\Delta t = 1/\sqrt{N}$ . Here  $0 \leq T \leq 5$  and  $M$  varies.

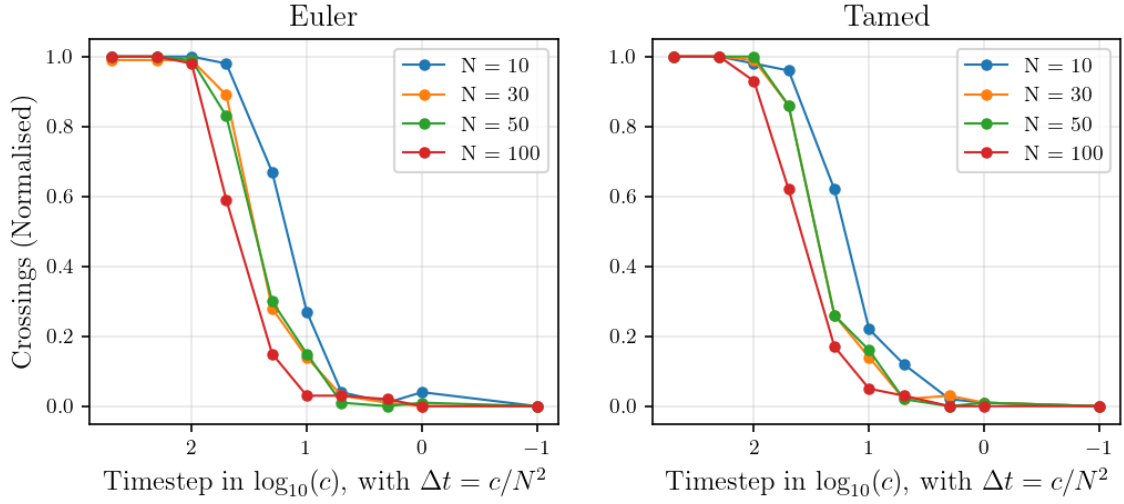


Figure 5: An old crossing experiment (since corrupted file). The constant  $c$  ranges from 500 to 0.1, and here  $M = 100$  trials are taken.

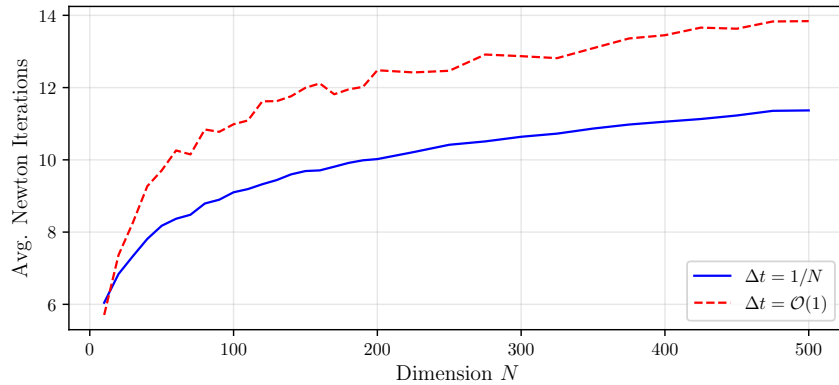


Figure 6: The average number of Newton iterations needed for the solve in the implicit method: averaged over  $M = 20$  trials, ran until  $T = 1$ , constant step size takes  $\Delta t = 0.1$ .